

NovaByte Solutions — Task 1

Data Exploration & Problem Framing Exploration Document

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Business Analytics with GenAI — Career 247 Capstone

1. Column Inventory

All five datasets, cross-checked against the brief's Data Dictionary (Section 9) and verified directly against the actual files.

Dataset	Actual rows	Columns	Main purpose
Clients	315	11	Customer/account master
Sales_Deals	2,540	13	Sales pipeline & revenue
Marketing_Campaigns	158	17	Marketing performance
Support_Tickets	4,060	14	Customer support
Finance_Expenses	1,025	12	Expenses & budgets

Clients.csv (315 rows, 11 columns)

Column Name	Data Type	Description
Client_ID	Text/String	Unique client identifier (CLI-XXXX); primary key, links to Sales_Deals and Support_Tickets.
Company_Name	Text/String	Name of the client organization.
Industry	Text/String	Industry vertical the client belongs to (Banking, Healthcare, Retail, Telecom, Manufacturing, etc.).
Contract_Type	Text/String	Type of engagement (Annual License, Multi-Year, Monthly SaaS, Project-Based, Managed Services).
Region	Text/String	Geographic region the client is based in (North, South, East, West).
City	Text/String	City of the client's office.
Onboarding_Date	Date	Date the client was onboarded; formats are mixed across rows.
Satisfaction_Score	Float	Client satisfaction rating, 2.5–5.0 scale.
Tenure_Years	Float	Years since onboarding.
Annual_Contract_Value	Integer	Annual contract value in INR.
Account_Status	Text/String	Current status: Active, Churned, or At Risk.

Sales_Deals.csv (2,540 rows, 13 columns)

Column Name	Data Type	Description
Deal_ID	Text/String	Unique deal identifier (DEAL-XXXXX); primary key for this table.
Sales_Rep	Text/String	Name of the sales representative handling the deal.

Column Name	Data Type	Description
Client_ID	Text/String	Foreign key linking to Clients.csv.
Product	Text/String	Product or service being sold.
Lead_Source	Text/String	How the lead originated (Google Ads, Referral, Webinar, etc.).
Deal_Stage	Text/String	Pipeline stage — Prospecting through Closed Won/Lost.
Deal_Value_INR	Integer	Monetary value of the deal in INR.
Created_Date	Date	Date the deal was created; formats are mixed.
Close_Date	Date	Date the deal closed; blank if still open.
Days_to_Close	Integer	Days from creation to closure; blank if still open.
Discount_Percent	Float	Discount offered on the deal.
Region	Text/String	Region the deal was sourced from.
Quarter	Text/String	Quarter and year the deal was created (e.g. Q1-2024).

Marketing_Campaigns.csv (158 rows, 17 columns)

Column Name	Data Type	Description
Campaign_ID	Text/String	Unique campaign identifier (CAMP-XXXX).
Campaign_Name	Text/String	Descriptive name of the campaign.
Channel	Text/String	Marketing channel used (Google Ads, LinkedIn, Email, etc.).
Campaign_Type	Text/String	Campaign category (Lead Generation, Brand Awareness, Event Promotion, Product Launch, Retargeting).
Start_Date	Date	Campaign start date; formats are mixed.
End_Date	Date	Campaign end date; formats are mixed.
Budget_INR	Integer	Total campaign budget in INR.
Impressions	Integer	Total ad impressions.
Clicks	Integer	Total clicks generated.
MQLs	Integer	Marketing Qualified Leads generated.
SQLs	Integer	Sales Qualified Leads generated.
Conversions	Integer	Number of deal closures attributed to the campaign.
CTR_Percent	Float	Click-through rate, as a percentage.
Cost_Per_Click	Float	Average cost per click in INR.
Customer_Acquisition_Cost	Float	Cost per acquired customer in INR; blank where there were no conversions.

Column Name	Data Type	Description
Target_Region	Text/String	Region the campaign targeted.
Target_Industry	Text/String	Industry the campaign targeted.

Support_Tickets.csv (4,060 rows, 14 columns)

Column Name	Data Type	Description
Ticket_ID	Text/String	Unique ticket identifier (TKT-XXXXX).
Client_ID	Text/String	Foreign key linking to Clients.csv.
Product	Text/String	Product the ticket relates to.
Issue_Type	Text/String	Category of the issue reported.
Priority	Text/String	Ticket priority: Critical, High, Medium, Low.
Created_Date	Date	Date the ticket was created; formats are mixed.
Resolved_Date	Date	Date the ticket was resolved; blank if unresolved.
Resolution_Hours	Float	Hours taken to resolve the ticket; blank if unresolved.
SLA_Target_Hours	Integer	SLA target resolution time in hours, based on priority.
SLA_Met	Text/String (Yes/No)	Whether the SLA was met.
Status	Text/String	Current status: Resolved, Open, In Progress, Escalated.
Assigned_Agent	Text/String	Name of the support agent assigned.
Contact_Channel	Text/String	How the client reached support (Email, Phone, Chat, Portal).
Customer_Satisfaction	Integer	Post-resolution satisfaction rating, 1–5.

Finance_Expenses.csv (1,025 rows, 12 columns)

Column Name	Data Type	Description
Expense_ID	Text/String	Unique expense identifier (EXP-XXXXX).
Department	Text/String	Department that incurred the expense.
Category	Text/String	Expense category (Software Licenses, Cloud Infrastructure, etc.).
Vendor	Text/String	Vendor or service provider name.
Amount_INR	Integer	Expense amount in INR; some rows are negative.
Budget_Allocated_INR	Float	Budget allocated for the expense.
Variance_INR	Float	Actual vs. allocated difference; negative means over budget.

Column Name	Data Type	Description
Expense_Date	Date	Date of the expense; formats are mixed.
Approval_Status	Text/String	Approved, Pending, or Rejected.
Payment_Mode	Text/String	Bank Transfer, Credit Card, Petty Cash, or Vendor Credit.
Fiscal_Year	Text/String	Indian fiscal year (e.g. FY2024-2025).
Remarks	Text/String	Optional free-text notes; blank in most rows.

2. Data Quality Issues

Found by directly scanning each file — full-row duplicate counts, null counts per column, date-format signatures, and 3σ outlier checks.

Clients.csv

Issue Found	Where	Potential Impact on Analysis
15 duplicate Client_IDs	Client_ID column	Will double-count that client's deals/tickets/revenue if not de-duplicated before any join.
Missing values in 4 columns	Company_Name (4), Industry (10), City (8), Satisfaction_Score (14)	Missing Satisfaction_Score values reduce the observations available for satisfaction analysis and may introduce bias if missingness is not random.
Mixed date formats	Onboarding_Date — 306 rows YYYY-MM-DD, 9 rows DD/MM/YYYY-style	Tenure_Years calculated from a mis-parsed date will be off by months or years for those 9 rows.

Sales_Deals.csv

Issue Found	Where	Potential Impact on Analysis
40 full-row duplicate deals	Whole table	Inflates pipeline volume and Deal_Value_INR totals if summed before de-duplication.
Missing values across 5 columns	Sales_Rep (50), Lead_Source (79), Discount_Percent (600); Close_Date & Days_to_Close (1,539 each — expected, these are still-open deals)	Sales_Rep/Lead_Source gaps will undercount rep- or channel-level win rates unless flagged rather than dropped.
Mixed date formats	Created_Date — 2,390 rows YYYY-MM-DD vs 150 DD/MM or MM/DD/YYYY; Close_Date split further across YYYY-MM-DD (946) and DD-MM-YYYY (55)	Quarter grouping and days-to-close math will be wrong for the non-standard rows unless dates are normalized first.
41 Deal_Value_INR outliers ($>3\sigma$ above mean)	Deal_Value_INR column — mean $\approx$ ₹2.79L, largest $\approx$ ₹19.8L	Could be legitimate large enterprise deals rather than errors — worth flagging and reviewing individually, not dropping outright.

Marketing_Campaigns.csv

Issue Found	Where	Potential Impact on Analysis
8 full-row duplicate campaigns	Whole table	Inflates total spend and MQL/conversion totals if not removed first.
Missing values	Impressions (7), Clicks (7), Target_Region (8), Target_Industry (15)	Target_Region gaps will undercount any region-level marketing rollup.
Channel vocabulary doesn't match Sales_Deals.Lead_Source	Channel column (8 distinct values) vs. Lead_Source (8 distinct values) — only 3 match exactly	Any join between the two tables on these fields will silently drop or mismatch rows unless mapped first (see Section 3).

Support_Tickets.csv

Issue Found	Where	Potential Impact on Analysis
60 full-row duplicate tickets	Whole table	Inflates ticket-volume and SLA-compliance denominators if not de-duplicated first.
Issue_Type has heavy casing/spacing inconsistency	40 distinct raw values collapse to 12 once whitespace/case are normalized (e.g. "API Integration Failure" vs "API Integration Failure")	Fragments a 'top issue types' analysis into near-duplicate buckets, hiding the real top offenders.
Missing values	Priority (83), SLA_Met (120), Assigned_Agent (131); Resolved_Date/Resolution_Hours (1,629 each — expected, open tickets); Customer_Satisfaction (2,005 — opt-in, post-resolution only)	SLA_Met gaps will understate the true breach rate unless explicitly excluded from the compliance denominator.
38 Resolution_Hours outliers ($>3\sigma$ above mean)	Resolution_Hours column — slowest ticket $\approx$ 120 hours	Worth a closer look at Issue_Type/Priority for these — may point to a specific bottleneck rather than random noise.

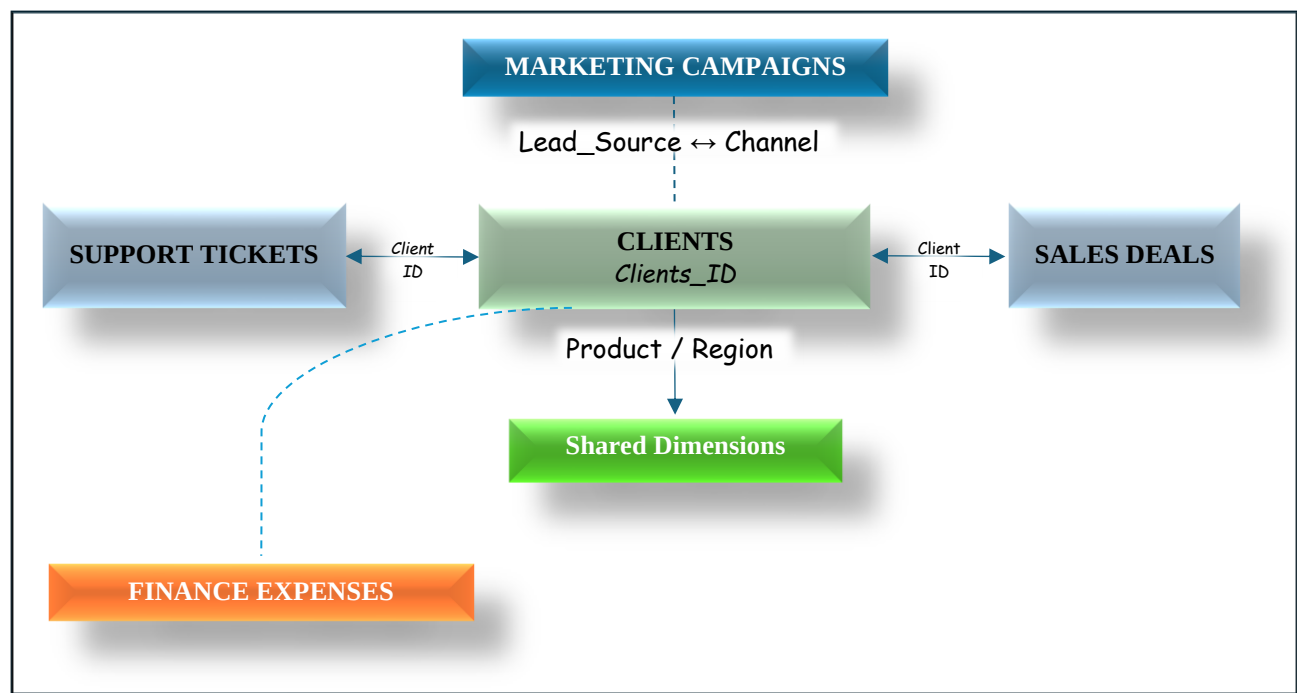
Finance_Expenses.csv

Issue Found	Where	Potential Impact on Analysis
25 full-row duplicate expense entries	Whole table	Inflates department/category spend totals if not removed first.
Missing values	Department (42), Category (24), Budget_Allocated_INR & Variance_INR (48 each)	A missing Department blocks that row from any department-level rollup — matches the brief's own note about vendors with unclear ownership.

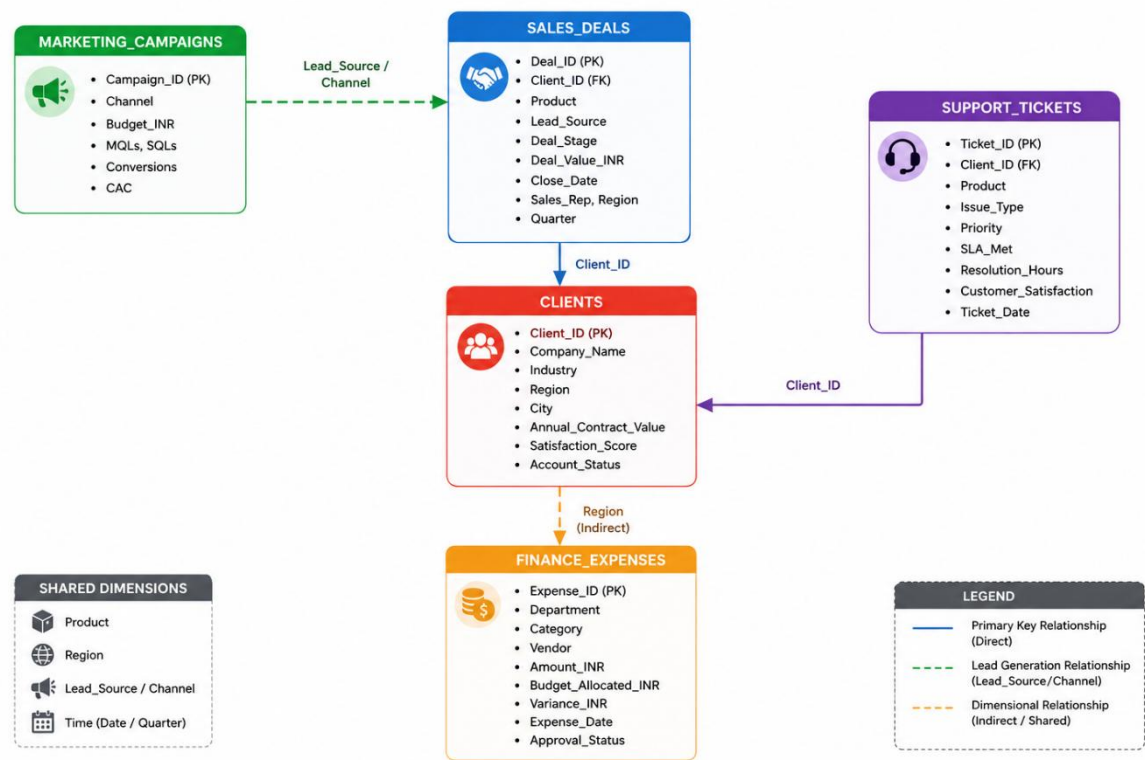
Issue Found	Where	Potential Impact on Analysis
40 negative Amount_INR values	Amount_INR column	Per the brief's hint, needs a judgment call — likely credits/refunds rather than entry errors — before deciding whether to keep, flag, or exclude.
Mixed date formats	Expense_Date — 968 rows YYYY-MM-DD vs 57 rows DD/MM or MM/DD/YYYY	Fiscal-year and monthly rollups will misclassify the 57 non-standard rows unless normalized first.

3. Relationship Map

Client_ID is the backbone — it's the only true foreign key linking Clients to both Sales_Deals and Support_Tickets. Region and Product are usable shared dimensions rather than formal keys. Finance_Expenses stands apart: it carries no Region or Client_ID field, only Department, so it has to be reported alongside the other four rather than joined directly.



NovaByte Solutions – Cross-Functional Relationship Diagram



Lead_Source → Channel mapping

The two tables describe the same underlying idea — where a lead came from — using different vocabularies. Only 3 of 8 values match exactly, so the rest need an explicit mapping before joining Sales_Deals to Marketing_Campaigns:

Lead_Source (Sales_Deals)	Mapped Channel	Note
Google Ads	Google Ads	Exact match
Trade Show	Trade Show	Exact match
Webinar	Webinar	Exact match
Email Campaign	Email	Same concept, different label
LinkedIn Campaign	LinkedIn	Same concept, different label
Organic Search	SEO	Same concept, different label
Cold Outreach	— no Channel equivalent	Direct sales outreach; not a tracked Marketing channel
Referral	— no Channel equivalent	Word-of-mouth / existing-client referral; not paid or campaign-driven

One thing worth flagging rather than assuming an answer to: Marketing_Campaigns also tracks two channels — Content Marketing and Social Media — that never appear as a matching Lead_Source. That could mean those channels drive awareness without producing a distinctly-tagged closed deal, or that Sales isn't tagging them consistently. Worth a follow-up question to the Marketing/Sales teams rather than guessing in the analysis.

4. KPI Definitions

Three KPIs per department, each with an explicit formula rather than just a name.

Sales

KPI	Formula	Why it matters
Win Rate	$\text{Closed Won} \div (\text{Closed Won} + \text{Closed Lost})$	Shows how much of the closed pipeline actually converts — the brief's core question is whether a 'healthy' pipeline is really healthy.
Average Deal Size	$\Sigma \text{Deal_Value_INR} (\text{Closed Won}) \div \text{Count of Closed Won deals}$	Flags whether deals are shrinking even while deal count looks stable.
Average Days to Close	$\text{AVERAGE}(\text{Days_to_Close})$ for closed deals, by quarter	Tracks whether the sales cycle is lengthening over time.

Marketing

KPI	Formula	Why it matters
Customer Acquisition Cost (CAC)	$\text{Total Spend} \div \text{Conversions}$	What it actually costs to win a customer, per channel.
MQL-to-SQL Conversion Rate	$\text{SQLs} \div \text{MQLs}$	Measures lead quality, not just lead volume — directly tests whether the 20% YoY spend increase is buying better leads.
Campaign ROI	$\text{Conversions} \div \text{Budget_INR}$	Flags which campaign types are worth the spend and which aren't.

Support

KPI	Formula	Why it matters
SLA Compliance Rate	$\text{SLA Compliance Rate} = \text{Tickets where SLA_Met} = \text{"Yes"} \div \text{Tickets with a known SLA_Met value}$	Directly tracks the service-quality slippage leadership flagged.
Average Resolution Time	$\text{AVERAGE}(\text{Resolution_Hours})$	Tracks whether support is getting slower, and for which issue types.
Escalation Rate	$\text{Tickets with Status} = \text{Escalated} \div \text{Total Tickets}$	Directly tracks the 'spike in escalations' symptom named in the brief.

Finance

KPI	Formula	Why it matters
Budget Variance	Allocated Budget - Actual Spend	Flags departments consistently running over budget.
Cost per Department	Σ Amount_INR, grouped by Department	Shows where budget is actually concentrated.
Rejected/Pending Rate	$(\text{Rejected} + \text{Pending}) \text{ Expenses} \div \text{Total Expenses}$	Flags approval friction — ties to the brief's note about unclear ROI on vendor contracts.

5. One-Page Analytics Plan

Question 1: Is the “healthy” sales pipeline actually converting at a healthy rate, or is win rate / deal velocity quietly declining behind steady pipeline volume?

Datasets needed: Sales_Deals.csv (primary), Clients.csv (context)

Answered in: Task 2 (Excel — win rate by rep/region, days-to-close), reinforced in Task 4 (H1 — channel win-rate test, H2 — closure-time t-test)

Question 2: Is the ~20% YoY jump in marketing spend buying better leads, or is cost-per-acquisition rising while lead-to-close conversion stays flat?

Datasets needed: Marketing_Campaigns.csv (primary), Sales_Deals.csv (joined via the Lead_Source ↔ Channel mapping in Section 3)

Answered in: Task 3 (SQL — spend/CAC/ROI by channel, closed-deal value attributed to each channel), Task 4 (H4 — quarterly spend vs. conversion trend)

Question 3: Is the drop in client satisfaction concentrated among clients with more support escalations — and are those the same clients closing fewer repeat deals?

Datasets needed: Support_Tickets.csv (primary), Clients.csv, Sales_Deals.csv

Answered in: Task 3 (SQL — SLA compliance, top-ticket-volume clients, cross-functional join), Task 4 (H3 — satisfaction vs. SLA-breach correlation)

Question 4: Where exactly are Finance's budget overruns concentrated, and which vendors or categories show high spend with no clear tie to results?

Datasets needed: Finance_Expenses.csv (primary)

Answered in: Task 2 (Excel — budget variance by department, top 5 vendors, flagged pending/rejected entries above ₹200,000)

Question 5: Once revenue, marketing efficiency, SLA compliance, and spend discipline are looked at together, which region is healthiest vs. weakest — and does that pattern explain part of the overall stagnation?

Datasets needed: Sales_Deals.csv, Marketing_Campaigns.csv (Target_Region), Support_Tickets.csv (via Clients.Region); Finance_Expenses.csv reported alongside (department-level only, no Region field)

Answered in: Task 3 (SQL regional rollup), Task 5 (dashboard's cross-functional view)